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Task Description:	Progress was made on the following tasks: 1) functional evidence for the HU-induced changes in selected signaling pathways, 2) evidence from Western analyses the HU may induce a change in vascular smooth muscle cells from a contractile to a synthetic phenotype, 3) functional evidence for HU-induced changes in RhoA/Rho kinase signaling 4) functional evidence for HU-induced structural changes in both carotid and femoral arteries, 5) HU-induced enhancement of vasoconstriction in endothelium-denuded but not endothelium-intact thoracic aorta.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:			

Task Progress:

In the following text, actual experimental observations are in upper case. Sections are headed by titles of each task completed.
FUNCTIONAL EVIDENCE FOR THE HU-INDUCED CHANGES IN SELECTED SIGNALING PATHWAYS

The goal of these experiments was to identify possible signaling pathways in vascular smooth muscle that could be altered by HU. In a previous preliminary study [1], the relatively nonselective tyrosine kinase inhibitor, genistein, inhibited the aorta contraction to norepinephrine (NE) in control, but not HU vessel rings. THIS EXPERIMENT WAS REPEATED IN THE PRESENT STUDY WITH SIMILAR RESULTS. Likewise, inhibition of cyclooxygenase with indomethacin exhibited the same differential pattern of contractile inhibition in control versus HU rings [1] AND THIS PATTERN WAS RETAINED AFTER ENDOTHELIAL DENUDATION. In contrast, the phospholipase C (PLC) inhibitor NCDC, was found previously to inhibit the NE-stimulated contraction equally in control and HU aorta rings [1]. This suggests that the signaling pathways associated with PLC remained functional after HU. This was explored further by assessing the effect of protein kinase C (PKC) blockade. PKC is activated by diacylglycerol, a metabolic product of PLC activity. BLOCKADE OF PKC WITH GF109203x MARKEDLY REDUCED NE-STIMULATED CONTRACTIONS NEARLY EQUALLY IN CONTROL AND HU AORTA RINGS. This agrees with our earlier observation using a structurally and mechanistically different PKC inhibitor, Calphostin C [2].

Nonselective blockade of tyrosine kinases markedly reduce the NE-stimulated contraction in control abdominal aorta [1]. This implies that one or more tyrosine kinases mediate NE-stimulated vasoconstriction. In contrast, tyrosine kinase blockade had no effect on the NE-stimulated contraction of HU aorta [1]. This implies that simulated microgravity downregulates selected tyrosine kinases and that this could contribute to the HU-induced impairment of vasoconstriction. Thus, experiments were carried out to identify the specific tyrosine kinases and/or tyrosine kinase dependent signaling pathways altered by HU. Src is a candidate tyrosine kinase because it has been shown to mediate vasoconstriction [3, 4]. BLOCKADE OF SRC WITH PP1 [5] MARKEDLY INHIBITED THE NE-STIMULATED CONTRACTION OF CONTROL, BUT NOT HU ABDOMINAL AORTA. p38 mitogen-activated protein kinase (p38 MAPK) and p42/44 MAPK, also called extracellular receptor regulated kinase (ERK1/2) are also candidate kinases for downregulation by HU because each has been shown to mediate vasoconstriction [6, 7] and to require tyrosine phosphorylation for activation [8]. BLOCKADE OF p38 MAPK WITH SB203580 MARKEDLY INHIBITED THE NE-STIMULATED CONTRACTION OF CONTROL, BUT NOT HU AORTA. In contrast, BLOCKADE OF THE ACTIVATION OF ERK1/2 BY INHIBITING MAP KINASE KINASE (MEK) WITH U0126 MARKEDLY AND EQUALLY INHIBITED THE NE-STIMULATED CONTRACTION OF BOTH CONTROL AND HU AORTA RINGS.

If HU downregulates src, p38 MAPK and/or ERK1/2, it could do so by either decreasing total protein expression or preventing the activation of these kinases. Experiments were carried out to address the first of these possibilities. WESTERN ANALYSIS REVEALED THAT HU CAUSED MARKED REDUCTIONS IN TISSUE LEVELS OF SRC AND p38 MAPK. IN CONTRAST, HU MARKEDLY ELEVATED TISSUE LEVELS OF ERK1/2.

In conclusion, HU treatment appears to downregulate src and p38 MAPK, and this may contribute to the HU-induced vasoconstrictor hyporesponsiveness. In contrast, any effects of HU on PLC, PKC or ERK1/2 proteins have little or no effect on contractile function.

EVIDENCE FROM WESTERN ANALYSES THE HU MAY INDUCE A CHANGE IN VASCULAR SMOOTH MUSCLE CELLS FROM A CONTRACTILE TO A SYNTHETIC PHENOTYPE

Studies of the abdominal aorta show that HU treatment yields a marked reduction in contractile capacity [9, 10]. Papadopoulus and Delp [11]demonstrated clearly that structural and morphological changes do not occur. Therefore, the mechanisms for the HU-induced vasoconstrictor hyporesponsiveness of this vessel are at a molecular level involving membrane receptors, signaling mechanisms and/or changes in the contractile apparatus. Western analysis was used to explore HU effects on contractile proteins of abdominal aorta. HU TREATMENT CAUSED MARKED REDUCTIONS IN alpha ACTIN, MYOSIN HEAVY CHAIN (MHC) AND MYOSIN LIGHT CHAIN 20 (MLC20). This profile is also observed in vascular smooth muscle cells in tissue culture. After a limited number of passages, these cells dedifferentiate [12]; that is, they transform from a contractile to a synthetic phenotype, characterized, in part, by a decline in contractile protein expression. In order to pursue the possibility of phenotype change, an additional marker protein, heavy caldesmon, was assessed. Heavy caldesmon is an immature form and is elevated in dedifferentiated vascular smooth muscle cells. HU TREATMENT INCREASES HEAVY CALDESMON. Preliminary results are also available from two experiments with an additional marker protein, calponin. Calponin expression is elevated in mature contractile cells. HU TREATMENT DECREASES CALPONIN EXPRESSION.

In conclusion, this set of experiments shows that HU treatment decreases contractile protein expression, possibly by inducing vascular smooth muscle dedifferentiation.

Additional measurements of contractile proteins were made in carotid and femoral arteries to determine if the pattern of the aorta was generalized. HU MARKEDLY REDUCED MLC20, BUT HAD NO EFFECT ON EITHER MHC OR alpha ACTIN IN CAROTID ARTERY. IN FEMORAL ARTERY, HU REDUCED BOTH MHC AND MLC20, BUT HAD NO EFFECT ON alpha ACTIN. In conclusion, HU has differential effects on the three arteries. However, MLC20 is reduced by HU in all three. Given the decisive role of MLC20 in mediating contraction, this change alone could contribute to the vascular hyporesponsiveness of the vessels.

FUNCTIONAL EVIDENCE FOR HU-INDUCED CHANGES IN RHOA/RHO KINASE SIGNALING

Western analysis was used to measure RhoA in abdominal aorta. HU TREATMENT MARKEDLY REDUCES RHOA EXPRESSION. In preliminary experiments, control and HU aorta rings were precontracted with UK 14,304, an alpha 2 agonist. Then the Rho kinase inhibitor, Y27632, was added increasing concentrations and relaxation measured. At 10 microM, Y27632 caused more than 80% relaxation in both control and HU artery rings. However, AT 3 microM, Y27632 CAUSED 100% RELAXATION IN THE CONTROL, BUT LESS THAN 25% RELAXATION IN HU. This suggests that the HU-induced loss of RhoA decreases the contribution of the RhoA/Rho kinase pathway to contraction. In turn, this could contribute the HU-induced vascular hyporesponsiveness. It was also found that HU decreases RhoA expression in carotid, but not femoral arteries (data not shown).

FUNCTIONAL EVIDENCE FOR HU-INDUCED STRUCTURAL CHANGES IN BOTH CAROTID AND FEMORAL ARTERIES

Structural changes in blood vessels can occur as a consequence of the hemodynamic effects of HU. Because the carotid

and femoral arteries experience, respectively, increases and decreases in both flow and pressure, passive and active length tension relationships were determined in order to detect any possible functional consequences that would occur if there were HU-induced structural changes. IN CAROTID ARTERY, THE HU EXHIBITED GREATER PASSIVE FORCES AT ALL LENGTHS GREATER THAN 0.4 mm; I.E., BETWEEN 0.6 AND 1.2 mm. This demonstrates that HU treatment caused the carotid artery to remodel by increasing stiffness. Interestingly, this did not change the position of the active curve on the X axis. THERE WAS NO HU EFFECT ON ACTIVE FORCE DEVELOPMENT IN THESE ENDOTHELIUM-DENUDED VESSELS, in agreement with our earlier findings [13].

IN FEMORAL ARTERY, HU DID NOT HAVE A SIGNIFICANT EFFECT ON THE PASSIVE LENGTH-FORCE RELATIONSHIP. THERE WAS A MARKED HU EFFECT ON THE ACTIVE LENGTH-FORCE RELATIONSHIP, SIGNIFIED BY THE MARKED REDUCTION OF HU FORCE DEVELOPMENT ON THE Y AXIS. HOWEVER, THE CONTROL AND HU ACTIVE FORCE CURVES HAVE IDENTICAL POSITIONS ON THE X AXIS. This fact, plus the lack of HU effect on passive force, demonstrates the HU treatment did not cause a change in the material or structural properties of the femoral artery. In order to perform length-force experiments, it was necessary to apply a minimal force to the artery rings just sufficient to change their shape from tubular to two parallel sheets. The resulting length was the unstretched length (Lu). IN HU CAROTID ARTERY, MORE THAN TWICE THE FORCE WAS REQUIRED TO REACH Lu, COMPARED TO THAT IN THE CONTROL CAROTID ARTERY (22.3 vs. 8.3 mg). This is consistent with the passive length-force relationship in carotid artery which showed that the HU vessel is stiffer than control. In contrast, THE FORCE REQUIRED TO ACHIEVE Lu IN THE HU FEMORAL ARTERY WAS LESS THAN HALF THAT REQUIRED IN THE CONTROL FEMORAL ARTERY (8.0 vs. 19.0 mg). This suggests that HU made the femoral artery more compliant. However, the magnitude of this effect was small as there was no difference between control and HU femoral arteries in the passive length-force relationship.

HU-INDUCED ENHANCEMENT OF VASOCONSTRICTION IN ENDOTHELIUM-DENUDED BUT NOT ENDOTHELIUM-INTACT THORACIC AORTA

In 1993, Delp and his coworkers [9] published a seminal work establishing that HU treatment decreases vasoconstrictor responsiveness. The major focus of their study was on the abdominal aorta. However, they also demonstrated hyporesponsiveness in the thoracic aorta. We have studied the thoracic aorta and have found that there is no HU effect. We use Wistar rats while Delp and his coworkers use Sprague-Dawley rats. This could be the explanation. In the present experiments, THERE IS NO HU EFFECT IN ENDOTHELIUM-INTACT THORACIC AORTA. Interestingly, WHEN THE ENDOTHELIUM IS REMOVED, THE HU AORTA EXHIBITS A STRONGER CONTRACTION THAT THE CONTROL. Experiments were also carried out in abdominal aortas from the same animals and a strong HU effect (vasoconstrictor hyporesponsiveness) was seen, validating the experiments in the thoracic aorta (data not shown). We predict the following in Sprague-Dawley rat thoracic aorta: There will be an HU effect, as reported by Delp et al. [9]. However, after removal of the endothelium, we predict that the both control and HU aorta rings will contract more strongly and the difference between them will disappear.

References

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Bibliography Type:	Description: (Last Updated: 12/28/2007)
Abstracts for Journals and Proceedings	Hwang, S., Haddad, F., Huynh, J., Le, S.N. Nguyen, J., Purdy, R.E. "Simulated microgravity alters vascular alpha-actin, myosin light chain 20 and myosin heavy chain." Experimental Biology, April 2-5, 2005 San Diego, CA FASEB J. 19:A669, 2005. , Mar-2005